ETNA: EnTropy regularization for compressed Neural Avatar

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Abstract

The growing demand for realistic and controllable virtual humans in large-scale applications such as online multi-user environments, gaming, and shared immersive realities raises challenges for memory efficiency and bandwidth usage. While implicit neural representations (INRs) provide a powerful alternative to explicit mesh-based avatars, they often require large neural networks whose parameters are not optimized for compression, limiting scalability. We introduce an entropy regularization framework for neural implicit avatar representations that shapes the weight distribution during training, making it suitable for standard compression schemes. Our approach adds regularization losses, including a differentiable soft-entropy loss and a L1 loss between quantized and unquantized weights, enabling the network to minimize entropy while maintaining high-fidelity geometric reconstruction. Unlike Quantization-Aware Training (QAT), our method does not require a custom forward pass and is meant to be used jointly with post-training compression tools. Experiments on DFaust, including unseen identities and poses, demonstrate significant reductions in model size while preserving generative capabilities and show improvements over standard PTQ and QAT baselines. Entropy regularization enables robust compression at lower bitrates without sacrificing visual fidelity and reconstruction accuracy.

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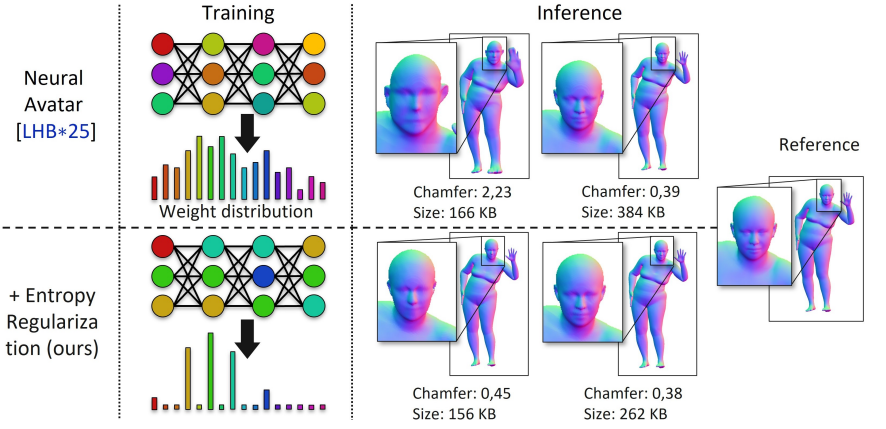


Figure 1: Method overview: Implicit virtual human models are trained with entropy regularization to reduce the encoded checkpoint size while minimizing the impact on visual quality. Top row: baseline model [22] trained without entropy regularization and its corresponding weight distribution histogram. Bottom row: model trained with entropy regularization. Inference: comparison of range-coded checkpoint size (kB) and reconstruction quality (Chamfer distance) between the two training settings. The proposed model outperforms the baseline both qualitatively and quantitatively. Both models are compressed with an entropy coding method, but our method preserves the visual quality at lower bitrates.

1 Introduction

Simulating realistic and controllable virtual humans is an active topic of research and development, leading to significant recent advances in both quality and computation time for digital human representations in immersive environments. However, comparatively less attention has been given to memory usage when dealing with large and complex models used in massively multi-user interactive applications. While memory consumption may not pose a major issue in applications where virtual human models are stored locally, such as in visual effects or offline video games, it becomes a critical factor in shared multi-user immersive environments, where large numbers of virtual human shapes and poses must be shared and transmitted efficiently. In such applications, memory requirements for high-quality virtual human simulation and rendering directly affect both the overall system memory footprint and the network bandwidth needed for transmission. Addressing this issue is also essential to mitigate the environmental impact of these technologies in terms of greenhouse gas emissions and energy consumption. Consequently, memory usage has a direct influence on 1) the responsiveness and scalability of shared immersive environments, and 2) the memory constraints of end-user devices.

The most common way to represent an avatar is through a mesh model, typically rigged with a skeletal structure to enable animation, and augmented with a texture model for rendering. Parametric models such as SMPL [21] and, more recently, ATLAS [28], or Anny [2] represent avatars using a compact set of learned parameters. SMPL essentially interpolates between known shapes and requires over five million parameters for the male and female templates (approximately 35 MB each). A unique model is needed locally, while a reduced set of parameters (e.g., ten β shape coefficients in the case of SMPL) suffices to select a

given identity. However, SMPL cannot easily accommodate new identities beyond the predefined β space, which can be an important limitation for personalized avatars. In addition, SMPL relies on a single geometric template with a fixed mesh topology, which constrains its ability to represent a wide range of detailed shapes.

To address these limitations, implicit neural representations (INRs) have emerged as a promising alternative, with demonstrated effectiveness for compressing images [1], videos [20], meshes [17], and point clouds [38]. INRs represent a signal with a neural network that maps spatial coordinates to signal values, such as occupancy or signed-distance values for 3D geometry. This continuous formulation can represent shapes at different levels of accuracy and with diverse topologies, and was initially explored for static 3D assets [27]. To animate realistic humans, previous works had to tackle additional constraints such as temporal continuity, generalization to diverse poses [11, 25] and to multi-identities [10, 22]. In this context, INRs have shown promise as generative avatar models that can be efficiently adapted to new identities [22]. Nevertheless, these methods are not explicitly designed to optimize the bit-level compressibility of the underlying neural network weights.

From a compression perspective, efficient encoding depends on identifying predictable patterns within the data. Entropy is directly related to this notion, as it measures the expected level of "surprise" in the outcomes of a random variable: frequent outcomes carry little surprise, while rare outcomes contribute more. Consequently, a signal with lower entropy requires fewer bits to encode than one with higher entropy. However, training a model with an entropy regularization can be unstable in practice due to the optimization of two different objectives (compression versus reconstruction) [1, 19, 36]. To mitigate this, we fixed most of the hyperparameters during training such as the quantization values, the weight boundaries, and the parameter σ used in the soft-entropy computation.

In this paper, we address the design of implicit virtual human representations whose parameters can be efficiently compressed while preserving 3D reconstruction quality, animation fidelity, and generalization. This setting differs from standard INR compression, where a neural representation is usually optimized for a single image, shape, video, or scene. While prior work has investigated INR-based compression for static 3D meshes [29, 31], these methods primarily target fixed geometry rather than generative implicit models conditioned on pose and identity. ETNA is an architecture-agnostic entropy regularizer. However, we focus on neural implicit avatars because they present distinctive challenges, including the representation and animation of highly non-linear human shapes, as well as generalization to unseen poses and identities. Compression therefore cannot be evaluated only by reconstructing a fixed geometry, but must also preserve the ability of the model to generalize under new conditions. After initial deployment, identities within the learned distribution can be specified using compact identity and pose parameters. ETNA targets the complementary cost of distributing and updating the shared model. This is also relevant for personalization, where a generic avatar model may be stored once and only the identity-specific fine-tuning residual needs to be transmitted.

ETNA targets the encoded size of model checkpoints for storage and transmission, rather than parameter count, runtime memory, or inference speed. Our experiments concern the geometric component of neural human models, we do not evaluate textures, hair, clothing appearance, or rendering.

We show that post-training quantization, as well as naive quantization-aware training, leads to visible artifacts even under mild compression. To overcome this issue, we introduce a differentiable soft-entropy regularization into the training process, based on soft histogram computation. The proposed method is evaluated on two state-of-the-art multi-identity im-

plicit models [22, 25], encoded with an off-the-shelf range coder. An overview of the method is shown in Figure 1. In summary, the main contributions of this work are as follows:

- We introduce ETNA, a non-invasive entropy regularization method for training neural implicit avatar models that reduces encoded checkpoint size with minimal impact on reconstruction quality.
- We propose a stable training strategy combining differentiable soft entropy, quantized/full-precision weight alignment, fixed weight boundaries, and stochastic weight averaging.
- We demonstrate experimentally that the proposed approach outperforms both Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) for two different neural avatar models.
- We show that the same regularization can compress identity-specific fine-tuning residuals, reducing the cost of personalizing a shared avatar model to new identities.

2 Related work

2.1 Neural avatars

In the last decade, implicit methods have offered compelling alternatives to traditional explicit mesh representations, especially for capturing details in complex 3D shapes such as virtual humans. Instead of storing vertex positions explicitly, implicit methods learn a continuous function representing the shape, typically as an occupancy field [23] or a signed distance function [27].

Although first explored on single rigid objects, several works extended implicit representations to animated avatars. The main challenge of representing characters in motion lies in the highly non-linear shape deformations and the lack of controllability of implicit methods. A common strategy is to predict the implicit shape in a canonical pose, before explicitly animating the mesh in the desired pose using traditional methods such as Linear Blend Skinning (LBS). NASA [14] predicts per-body-part occupancy conditioned on skeletal transformations to model pose-dependent deformations. However, using body parts often results in undesirable seams under unseen poses. COAP [25] improves these artifacts by conditioning the body parts on their neighboring parts, but does not fully solve the problem.

Predicting the shape in canonical space itself can also be difficult, as datasets often contain posed characters, without correspondences with the canonical space. Methods such as Neural-GIF [34] and LEAP [24] learn a mapping between pose and canonical space, but this mapping can be ambiguous (non-bijective correspondences) and challenging to extrapolate to extreme or unseen poses.

SNARF [9] and Fast-SNARF [11] start from a rough estimate of a point's canonical position and iteratively update it until it reaches the desired location in pose space. As in other approaches, they jointly learn LBS weights and the implicit field, but their forward skinning approach (canonical to posed) makes the mapping easier to optimize. More broadly, many neural avatar methods leverage a parametric model such as SMPL [21] for canonicalization, animation, or as a shape prior [8, 25]. This dependency can introduce additional deployment overhead (e.g., requiring access to SMPL at inference).

Several systems further generalize across identity and clothing. For example, gDNA [10] targets detailed generative avatars, SMPLicit [12] uses a generative model for clothed bodies,

and AutoAvatar [3] explores autoregressive fields for dynamic avatars. For avatar personalization from consumer sensors, works such as PINA [15], Avatar [37], and ANIM [30] reconstruct animatable avatars from limited RGB-D inputs. A recent survey provides a broader view of neural-field avatars [18].

Other works convert implicit neural avatar fields into hybrid explicit representations. For example, BakedAvatar [16] precomputes neural-field information and bakes it into an explicit mesh and texture representation. This transformation enables faster rendering. Such explicit and hybrid approaches are complementary to ETNA. ETNA preserves the original neural implicit avatar architecture and instead changes the distribution of its weights to improve entropy encoding.

2.2 Entropy Regularization

In our context, Shannon entropy provides a lower bound on the average number of bits per (typically quantized) parameter required by an optimal lossless coder [32]. Lower entropy therefore suggests smaller bitstreams. This view motivates training objectives that encourage low-entropy parameter distributions.

To make entropy terms trainable, two things are required: 1) a differentiable quantization and 2) soft histograms to compute probabilities. VQ-VAE [36] popularized the Straight Through Estimator (STE) for hard assignments, where gradients bypass the non-differentiable rounding/nearest-neighbor step during backpropagation. Another widely used alternative is to add uniform noise to pre-quantized variables [1, 26], which yields a differentiable surrogate and improves robustness to quantization.

Although not directly related to entropy, Histogram Loss [35] introduced soft-binned histograms with differentiable counting rules. They used the soft histograms to compute similarities between embeddings in metric learning problems.

Closer to our setting, HEMP (High-Order Entropy Minimization) [33] directly regularizes the entropy of quantized parameters by capturing dependencies between successive symbols. Soft-to-Hard VQ [1] replaces hard vector quantization with a soft assignment to learned codewords (softmax over distances). The soft assignment is progressively hardened using an attribute similar to the σ parameter used in our kernel function (see Section 3.3). The authors note that the annealing schedule is crucial and can lead to optimization issues if chosen poorly [1]. Related work in neural image compression reports potential instability in practice [19, 36].

A standard approach to improve quantized inference is Quantization Aware Training (QAT) [13], which inserts fake-quantization during the forward pass and uses STE to update full-precision weights. QAT typically outperforms Post-Training Quantization (PTQ), but it does not explicitly encourage a compressible weight distribution and often requires modifying the forward graph which increases architectural complexity. In contrast, our approach is non-invasive: it only requires access to a model's weights and produces differentiable losses (soft-entropy and quantization-alignment) that decrease the entropy of the model while maintaining high inference quality.

3 Method

In this section, we present our method for training neural implicit networks with entropy regularization to accurately reconstruct virtual humans from compact encoded checkpoints.

Section 3.1 presents the fundamentals of neural implicit avatars. Section 3.2 reviews the entropy usage in the context of compression. Section 3.3 describes the construction of the entropy loss in the neural implicit virtual human representation. Finally, section 3.4 details the training procedure of the model.

3.1 Neural implicit avatar network

Fast-SNARF [11] introduced a generative implicit method capable of encoding a single identity across a wide range of poses. During inference, points are first sampled in a canonical space. A neural network then predicts the occupancy of the character, while a second network estimates the corresponding linear blend skinning (LBS) weights. The predicted occupancy is converted into a mesh representation using the marching cubes algorithm, and the skinning weights are subsequently applied to animate the avatar in the target pose. In addition, the occupancy network is conditioned on a pose parameter, allowing the model to capture non-linear, pose-dependent deformations.

This approach has recently been extended to support multiple identities [22]. By conditioning both the occupancy and linear blend skinning networks on an identity parameter, the generative model can represent several distinct identities without increasing the size of the networks. This property constitutes an important step toward reducing transmission costs: only the trained model and the identity–pose parameters are required to generate a character, if this character is within the training distribution. COAP, another generative neural avatar model, represents humans with body parts. In this paper, we use these two state-of-the-art generative models as baselines. Our goal is to fine-tune pre-existing methods with an entropy regularization to make the resulting generative model more appropriate for compression while preserving reconstruction quality as much as possible. The entropy regularization encourages the generative model's internal weight distribution to become more compressible while preserving its ability to represent multiple identities.

3.2 Entropy Regularization

In information theory, Shannon entropy provides the theoretical minimum number of bits required to encode a message. It is defined in terms of the frequency of occurrence of each symbol in the signal, independently of their order. A message in which all symbols are equally present yields the highest entropy, since the same number of bits must be used for each symbol. Conversely, a message in which one or a few symbols are highly frequent has lower entropy. Intuitively, frequent symbols can be encoded with fewer bits, resulting in a smaller average message size. The entropy is formally expressed as $H = -\sum_{a \in A} p(a) \cdot \log_2(p(a))$, where $p(a)$ denotes the probability of observing symbol a in the alphabet A.

The function $-p(a)\log_2(p(a))$ is concave, non-negative on the interval $[0,1]$, and reaches zero when $p(a)$ equals 0 or 1. Minimal entropy is therefore achieved when a message consists of only a few unique symbols with probabilities close to one. Modern entropy coding methods can achieve compression rates that approach this theoretical bound. Our objective is thus to minimize the entropy of a generative model's weights in order to reduce the encoded size, without degrading reconstruction quality. This is challenging because lower entropy means less available information and thus lower expressiveness.

Following the same definition, we treat the weights of a neural network as the message to be encoded. This is a natural assumption in the case of implicit neural representations, which are typically trained on a server and transmitted to client devices. Computing entropy

directly over all weights is, however, impractical. Neural networks store weights as floating-point values, which can take millions of unique values, resulting in very high entropy. A simple first step to minimize the entropy with small effect on the quality is to apply a light quantization (e.g. $n = 10$ bits). To achieve this, all parameters across layers are concatenated into a single one-dimensional tensor, normalized to the range $[0, 2^n - 1]$, and rounded to the nearest integer (see Eq. 1). The frequency of each integer is then computed to estimate the model's entropy.

More formally, let the weights of a neural network be represented as a flattened tensor $W \in \mathbb{R}^T$, where T is the total number of parameters. Our baseline consists of two separate networks, but we treat their parameters jointly by concatenating them into a single tensor. Let W_{norm} denote the normalized weight tensor, rescaled to lie within $[0, 1]$. The quantized tensor of weights is then defined as:

$$W^q = \left(\lfloor w_{\text{norm}} \times (2^n - 1) \rceil \right)_{w_{\text{norm}} \in W_{\text{norm}}}, \tag{1}$$

where each element of W^q is an integer in the range $\{0, 1, \ldots, 2^n - 1\}$.

Let $\overline{W^q} = \text{set}(W^q)$ denote the set of unique quantized values observed in W^q. The entropy of the quantized model weights is then expressed as

$$H = -\sum_{\bar{w} \in \overline{W^q}} p(\bar{w}) \cdot \log_2(p(\bar{w})), \tag{2}$$

where $p(\bar{w})$ is the empirical probability of observing the value $\bar{w}$, estimated by counting its frequency in W^q.

3.3 Entropy loss construction

By adding H (see Eq. 2) to the generative model's training loss, the optimizer is encouraged to modify the distribution of its internal weights in order to minimize entropy and, consequently, reduce the model size after compression. However, several of the previously described steps are not differentiable, especially the rounding operation during quantization, and the counting operation used for the frequency estimation. The rounding step could in principle be approximated by a differentiable step function, but such a function is flat almost everywhere, leading to vanishing gradients. To address this issue, we adopt the Straight-Through Estimator (STE) [5], which performs the forward pass with rounding while treating the rounding as an identity during backpropagation. This approximation is sufficient for training since rounding does not significantly alter the underlying data distribution.

This strategy, however, cannot be directly applied to histogram computation. Once the weights are quantized, a histogram is constructed to estimate the probability that a given element $\bar{w}$ is present in W^q. Since histogram counting is non-differentiable, we approximate it with a soft-histogram. The histogram is divided into 2^n bins, one for each integer in the range $\{0, 1, \ldots, 2^n - 1\}$. Thus, the set of unique quantized values $\overline{W^q}$ has at most 2^n elements, although in practice fewer bins are occupied.

To compute the number of elements in a bin b, we evaluate the distance between b and all quantized weights $w^q \in W^q$. If w^q is assigned to bin b, the distance is zero; otherwise, it is at least one. To approximate this count in a differentiable manner, we introduce a kernel function $f(x) = e^{-\sigma x^2}$, where σ is a temperature parameter controlling the decay rate of f. This function maps 0 to 1 and maps non-zero distances toward 0. The number of

weights assigned to bin b can therefore be approximated as $C(b,W^q) = \sum_{w^q \in W^q} f(w^q - b)$. Normalizing by the total number of weights T yields an estimate of $p(b)$, the probability of selecting an element from bin b. The resulting soft entropy is given by

$$\tilde{H} = -\sum_b p(b) \cdot \log_2\big(p(b) + \varepsilon\big) = -\sum_b \frac{C(b,W^q)}{T} \cdot \log_2\left(\frac{C(b,W^q)}{T} + \varepsilon\right), \tag{3}$$

where T denotes the total number of model parameters, and ε is a small constant (set to 10^{-12}) for numerical stability.

To minimize entropy, we need to compute the gradient of $\tilde{H}$ with respect to the weights W. In particular, the gradient depends on $\frac{\partial f(x)}{\partial x} = -2\sigma x e^{-\sigma x^2}$. This derivative vanishes at $x = 0$. Large values of σ produce sharply peaked kernels, leading to vanishing gradients for $|x| \geq 1$. Conversely, small values of σ yield broader kernels, with non-negligible gradients across neighboring bins. In practice, we fix $\sigma = 0.1$, which provides stable and informative gradients during training.

One issue arising from not using quantized weights during the forward pass is the misalignment between the reconstruction loss and the entropy regularization. For example, we noticed during our experiments that the network may set two weights to extreme positive and negative values. While such extremes may have little effect on reconstruction (using unquantized weights), they strongly influence normalization, leading to wider bins. This consequently artificially reduces entropy by merging more values into a single bin, but at the cost of reducing expressiveness of the quantized weights. To mitigate this issue, following prior work [33], we add an L_1 penalty between the quantized and unquantized weights. Although helpful, the L_1 loss alone is insufficient. We therefore introduce a stricter constraint by imposing fixed weight boundaries. Specifically, we define two hyperparameters b_{min} and b_{max} such that all values outside $[b_{min}, b_{max}]$ are clamped to this range during inference. During training, an additional loss ensures that the minimum and maximum weights converge to those boundaries. Doing so fixes the quantization values, increasing the stability of the training. By not using quantized weights during the forward pass, our proposed regularization losses are straightforward to add to existing methods.

The overall training objective combines four terms: 1) the reconstruction loss, 2) the L_1 alignment loss between quantized and unquantized weights, 3) the boundary penalty enforcing the range $[b_{min}, b_{max}]$, and 4) the soft entropy regularization. The total loss is expressed as

$$\mathcal{L} = \lambda_R \mathcal{L}_R + \lambda_{L_1} \mathcal{L}_1(W,W^q) + \lambda_{\max} \mathcal{L}_{\max} + \lambda \tilde{H}, \tag{4}$$

with:

- $\mathcal{L}_R$: the reconstruction loss, implemented as a Binary Cross Entropy (BCE) between predicted and ground-truth occupancy [11, 22];
- $\mathcal{L}_1(W,W^q)$: the L_1 penalty enforcing similarity between unquantized weights W and quantized weights W^q;
- $\mathcal{L}_{\max}$: the boundary penalty, defined as the difference between b_{max}(/b_{min}) and the smooth maximum (/minimum) computed with log-sum-exp;
- $\tilde{H}$: the soft entropy regularization term defined in Eq. 3;

- $\lambda_R, \lambda_{L_1}, \lambda_{\max}, \lambda$: scalar coefficients balancing the contributions of the four terms. The same λ is used in the results for the soft entropy weight.

This formulation slightly reduces expressiveness but effectively prevents extreme weights, thereby stabilizing entropy regularization.

3.4 Training

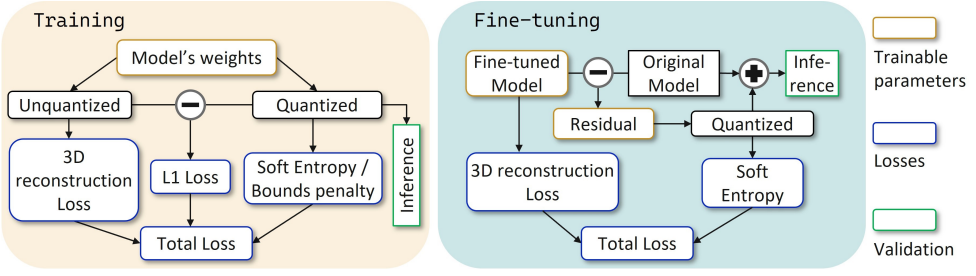


Figure 2: Block diagrams presenting a model's training and fine-tuning with the proposed entropy regularization.

Global overviews of the training and fine-tuning pipelines are shown in Figure 2. During the training, the unquantized weights are used for the 3D reconstruction task, while the quantized weights compute the soft entropy and boundary penalty. They are also used during inference. An additional L1 loss is taken between the unquantized and quantized weights. The model is first trained on the reconstruction task alone before gradually adding the entropy regularization halfway through the training. To mitigate the inherent instability of training with an entropy regularization, we applied Stochastic Weight Averaging (SWA). More details about the training procedure can be found in the supplementary materials.

LHB [22] proposed a fine-tuning approach to adapt generative implicit models to any new identity, even outside of the β space. Their method creates a frozen copy of the original model to supervise the fine-tuning in a teacher-student approach. We hypothesize that the new fine-tuned model might not be very different from the original model. We thus applied our entropy regularization to the residual between the original and fine-tuned models. In a real-world scenario, we could legitimately assume that a base generic model is present on local devices or a central server and that only the individually fine-tuned residual is transmitted.

4 Results

This section presents the results obtained by adding the entropy regularization to two state-of-the-art neural avatar models: COAP [25] and LHB [22]. These models were trained on the synthetic DFaust dataset and tested on unseen motions and identities. The quantitative metrics used are the Intersection over Union (IoU) of correctly predicted occupancy of points sampled on the surface and the Chamfer distance between the simulated and ground-truth meshes. Among all the metrics that we tested, the surface IoU is the most discriminative regarding high-frequency details, while the Chamfer distance is an intuitive global metric. More information about the dataset and metrics can be found in the supplementary materials.

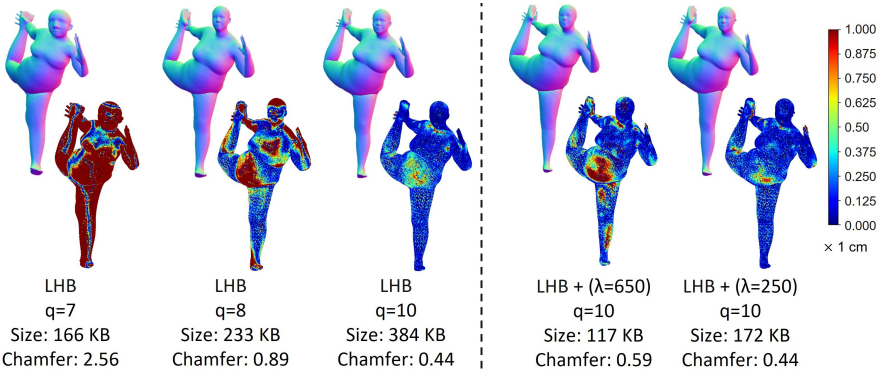


Figure 3: Meshes obtained with LHB for a pose unseen during the training. Left: model without entropy regularization. Right: model with entropy regularization for various soft entropy weights ($\lambda = 650$ or $\lambda = 250$). For each model, the figure reports the quantization level, the encoded file size after compression with range coder, and the mean chamfer distance in cm. The entropy regularization models perform better in low bitrate regimes.

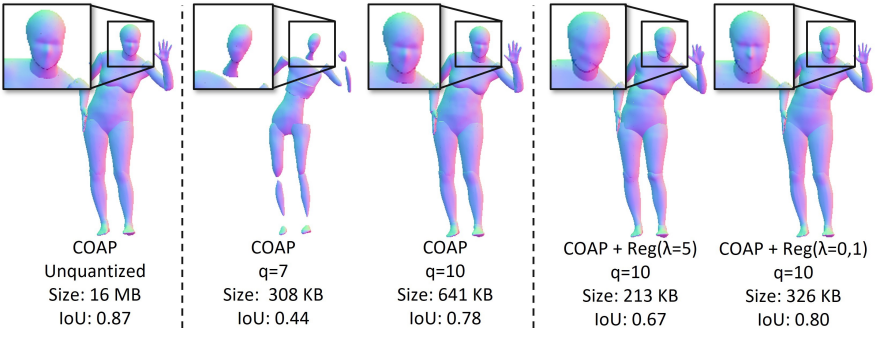


Figure 4: Meshes obtained with COAP for a pose unseen during the training. Left: model without entropy regularization. Right: model with entropy regularization for various soft entropy weights ($\lambda = 5$ or $\lambda = 0.1$). For each model, the figure reports the quantization level, the encoded file size after compression with range coder, and the IoU. The entropy regularization models perform better in low bitrate regimes.

4.1 Qualitative results

Figure 3 illustrates compressed reconstructions for a challenging unseen pose, not included in the training set (please see supplementary video for further results). The pose is taken from the *ulr1d* sequence of the MPI PosePrior dataset [2]. The checkpoint files are compressed using the lossless entropy encoder range coder. For the baseline without entropy regularization, the encoded size was reduced by lowering the quantization level q. For the proposed method, which incorporates entropy regularization, the compression rate was controlled through different values of λ (the weight of the soft entropy loss). This strategy provides finer control over the compression rate than adjusting the quantization level alone.

As shown in Figure 3, the proposed model (right) obtained a similar Chamfer distance to the non-regularized model (LHB [22] with $q = 10$) but with less than half the compressed

size (172 KB versus 384 KB). By contrast, to achieve similar compressed size, the baseline on the left-hand side of Figure 3 must be quantized to $q = 7$ bits, which drastically degrades the mesh quality. Although it sacrifices the high frequency details (on the face for instance), the regularization maintains a recognizable identity and human shape even at low bitrates (117 KB and 166 KB after compression for both proposed and baseline, respectively).

Figure 4 presents similar results for COAP: on the left is the model trained without entropy regularization and on the right trained with the regularization. The regularization halves the encoded size (326 KB versus 641 KB) with similar IoU (0.80 versus 0.78, respectively). To obtain a similar bitrate (around 300 KB), the unregularized version needs to be quantized to 7 bits, which significantly degrades the reconstruction quality. It is clear from Figures 3 and 4 that the proposed method has strong advantages in preserving the mesh identity structure at low bitrates compared to the naive baseline approaches.

	Total # of param (millions)	Entropy (bits/parameter)	Total bits (millions)	Chamfer Dist (cm)
SMPL [21] Female	5.22	20.00	104.35	N/A
gDNA [10]	31.13	24.45	761.25	N/A
COAP [25]	1.27	16.02	20.30	0.49
Fast-SNARF [11]	0.52	18.98	9.86	0.30
LHB [22]	0.52	18.99	9.92	0.30
COAP + Regularization	1.27	2.97	3.77	0.47
LHB + Regularization	0.52	2.40	1.25	0.30

Table 1: For each type of generative model, the total number of parameters, the entropy of the parameter distribution (average bits per parameter), the theoretical minimal number of bits needed to encode all parameters, and the Chamfer distance computed on the DFaust synthetic dataset. Our regularization reduces the entropy of LHB by approximately a factor of 8 while achieving similar Chamfer results.

4.2 Quantitative results

This section reports quantitative results and compares the reconstruction quality across various compression levels with state-of-the-art explicit and implicit models (see Table 1). Firstly, we present the number of parameters, the computed entropy and the theoretical compression rate. Finally, we report the rate-reconstruction trade-off using actual encoded sizes.

4.2.1 Model parameters and entropy

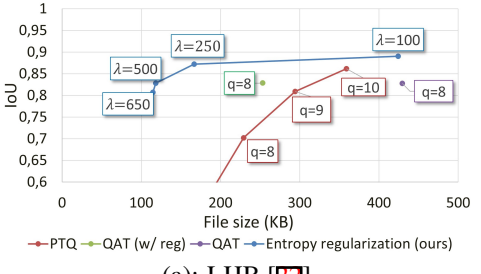
Table 1 compares different explicit and implicit model sizes. For each, the full size unquantized parameters are aggregated into a flattened tensor. The entropy is then computed from the histogram, yielding the minimum average number of bits required to encode each parameter. Multiplying this entropy by the total number of parameters provides a theoretical lower bound on the number of bits needed to encode the model. For SMPL, we use the parameters contained in the *SMPL_FEMALE.pkl* file, while the implicit models (gDNA [10], COAP [25], Fast-SNARF [11], and LHB [22]) are mostly composed of neural network weights. We apply our entropy regularization to COAP and LHB, as the baseline state-of-the-art on implicit multi-identity avatar models. Table 1 shows that without entropy

regularization, all methods exhibited an average of 16–24 bits per parameter. When using entropy regularization, this value can be reduced to approximately 2.4 bits per parameter, while maintaining an equivalent Chamfer distance. The Chamfer distance is computed with respect to an SMPL mesh, which is considered as ground truth. See the supplementary materials for a visual comparison of all implicit methods at low bitrates using different body shapes, e.g., male and female. Empirically, models with high entropy cannot be compressed drastically without a major impact on reconstruction quality.

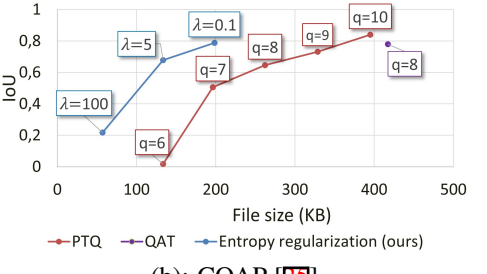
4.2.2 Rate-Reconstruction trade-off

Figure 5 presents the results obtained with our regularization versus standard compression techniques such as Post-Training Quantization (PTQ) and Quantization Aware Training (QAT). The figures plot the IoU against the file size in KB, where Figure 5(a) applies to the LHB and 5(b) to the COAP method. Results are computed on unseen identities and unseen poses i.e., entirely excluded from the training set. For the proposed method (blue curve), the results are obtained with different soft-entropy coefficient λ values and with a quantization of 10 bits. For the PTQ (red curve), the quantization level q was tuned to control the bitrate. We observe that both the IoU and the entropy increase along with q. For the QAT, we used PyTorch modules with a quantization of $q = 8$ bits. In both figures, QAT improves the IoU of the PTQ with $q = 8$, at the cost of a higher entropy. Indeed, as the model is trained with 8 bits quantization, to improve the reconstruction, it needs to increase the amount of information present in the weights, thus increasing the entropy. In Figure 5(a), we trained the model with both the entropy regularization and the quantized forward pass. The result demonstrates improvement over the PTQ, although less significant than the regularization alone. During the training, the L1 loss between quantized and unquantized weights is often around 2×10^{-5}, indicating that it is an adequate alternative to a quantized forward pass. On the other hand, PyTorch's QAT requires 8 bits quantization, thus limiting the regularization capabilities of the model.

Figure 6 reports the results obtained on the full DFaust dataset [6] composed of five male and five female subjects. IoU is computed on unseen identities and poses. The non-regularized model reaches an IoU of 0.84 with 9 bits quantization, corresponding to a file size of 331 KB. A comparable IoU is obtained with our regularized model at $\lambda = 250$, with a file size of 141 KB, thus dividing by around 2.3 the file size. The regularization seems more



(a): LHB [22]



(b): COAP [25]

Figure 5: Plot of the IoU versus the file size (in KB) for two different models: LHB [22] on the left and COAP [25] on the right. They compare the entropy-regularized training (blue line), QAT with regularization (green), QAT without regularization (purple), and PTQ (red line).

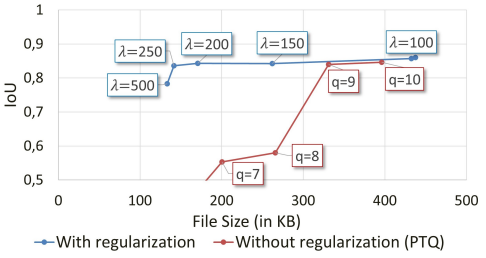


Figure 6: IoU with respect to file size in KB for our proposed regularization versus Post Training Quantization, on the full synthetic DFaust dataset

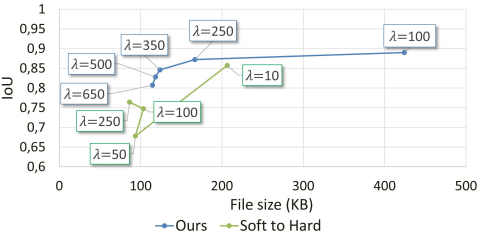
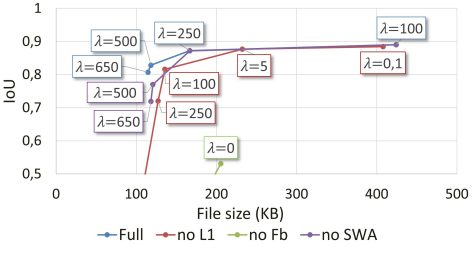
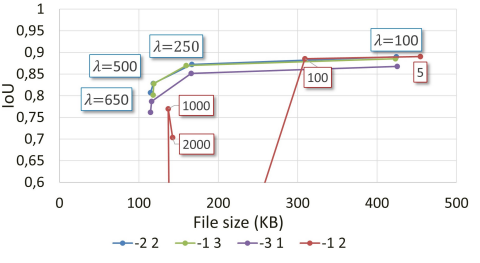


Figure 7: Comparison with Soft-to-Hard [1] applied on LHB. Soft-to-Hard can reach lower file sizes, but with higher reconstruction error



(a)



(b)

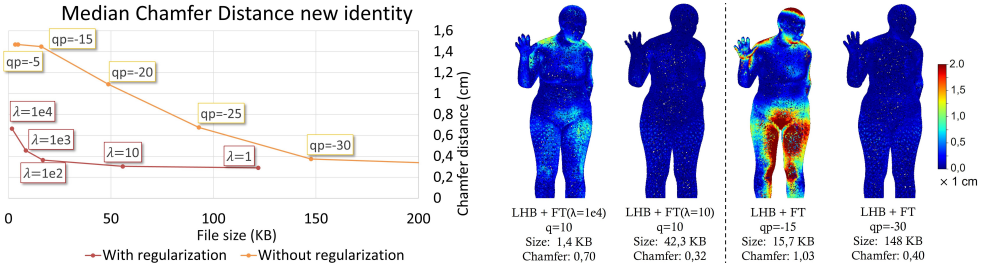
Figure 8: Parameter study of our proposed method applied to LHB. Figure 8(a) is the ablation study of our regularization method: without the L1 loss in red, without fixed boundaries penalty (no Fb) in green, and without the SWA in purple. Figure 8(b) shows variations of the weight boundaries.

impactful on the larger dataset, likely due to the increased difficulty of the reconstruction task.

Figure 7 compares ETNA with Soft-to-Hard [1], the closest entropy-regularized baseline. Since no official code was available, we reimplemented it as closely as possible to the original paper. Similar to other papers [19, 36], we observed that the Soft-to-Hard annealing schedule was sensitive in our setting and careful retuning was required. Even after retuning, ETNA performs better in the practically useful bitrate range, while Soft-to-Hard reaches slightly smaller files only at the cost of clearly higher reconstruction error. ETNA uses a fixed softness parameter and leaves the model forward pass unchanged, making it easier to integrate into existing neural implicit human-geometry pipelines.

4.2.3 Parameter study

Figure 8 studies the components and hyperparameters of our training strategy on the results. In particular, Figure 8(a) shows that training without the boundaries penalty completely breaks the regularization. Indeed, the model sets two weights to extreme values, making the quantized and unquantized weights misaligned. Removing either the L1 loss or Stochastic Weight Averaging (SWA) leads to degraded performance at low bitrates. However, they have minimal impact at higher bitrates. Figure 8(b) indicates that the model is stable for different fixed boundaries. For instance, clamping weights between $[-2,2]$ or $[-1,3]$ gives



(a) (b)

Figure 9: Residual compression when fine-tuning on a new identity: regularized version has varying λ and is encoded in 10 bits with range coder, unregularized version is compressed with NNCodec [4] for different qp values. Figure 9(a): entropy-regularized fine-tuning achieves a better rate-distortion trade-off than unregularized fine-tuning. Figure 9(b): error maps (in cm) illustrate that comparable or lower geometric error can be reached at substantially smaller residual bitstreams with regularization.

almost identical results; $[-3,1]$ is slightly below as the original weights approximately lie in $[-1.5,4]$. However, a model trained on a smaller interval such as $[-1,2]$ seems to be less reliable with two training runs failing to converge.

4.2.4 Regularized fine-tuning

Figure 9 demonstrates the gain in residual file size after fine-tuning LHB's generative model [22] with our entropy regularization, compared with naive post-training encoding. During the fine-tuning, the residual is regularized with different λ soft entropy weights, then quantized to 10 bits before being encoded with range coder. The non-regularized residual is compressed with NNCodec [4], a lossy encoder with a built-in quantization parameter qp enabling finer control over the reconstruction-bitrate trade-off. To have a representative sample, we ran five leave-one-subject-out experiments, each excluding one female subject from the original training set. The model is then fine-tuned on the missing character. We then selected the median results. We obtained similar results with the mean. Figure 9(b) shows fine-tuning on character 50022.

5 Conclusion

In this paper, we tackled efficient compression of implicit neural virtual human models. The main question we addressed is whether post-training quantization of existing networks is an effective strategy. We showed that it rapidly yields unacceptable visual reconstruction quality in moderate to low bitrate regimes. Instead, ETNA adds a non-invasive soft-entropy objective without modifying the model's forward pass. We show that the entropy weight can control the rate-reconstruction trade-off for a given quantization level. This yields a more efficient compression method, robust at lower bitrates with visually good reconstructions. Evaluation is currently limited to two occupancy-based models and primarily synthetic DFaust data.

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